

# README: ComStock 2025 Release 3

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## Introduction

This README provides key information to help users navigate and understand the ComStock dataset. The commercial building sector stock model, or ComStock™, is a highly granular, bottom-up model that uses multiple data sources, statistical sampling methods, and advanced building energy simulations to estimate the annual subhourly energy consumption of the commercial building stock across the United States.

This document outlines the structure and contents of the ComStock dataset on the Open Energy Data Initiative (OEDI) data lake, describes the file types and how to open them, explains naming conventions, and highlights recent changes and additions, including new upgrade measures and updates to the baseline building stock model. It also provides guidance for properly citing the dataset.

# OEDI Data Lake Contents

The contents of the OEDI data lake are summarized in the table, below.

| Name                                   | Contents                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| building_energy_models                 | Building energy models, in <a href="#">OpenStudio</a> format, that were run to create the dataset. Note that OpenStudio files can be converted to EnergyPlus IDF files as needed.                                                                                                                                                           |
| commercial_gap_model                   | Estimates the county-level, hourly electricity consumption of Commercial-sector (building and non-building, e.g. street lighting) uses not modeled in ComStock – the “commercial gap”. See Section 5.6 in the <a href="#">ComStock reference documentation</a> for more detail.                                                             |
| comparison_plots                       | Plots comparing this ComStock release to CBECS 2018, including energy consumption, and energy use intensity by fuel type, building type, vintage, and census division, and floor area by building type, vintage, and census division.                                                                                                       |
| component_loads                        | Files and plots that disaggregate the causes of heating and cooling in ComStock buildings. For more information, please refer to the <a href="#">Commercial Building Load Components</a> explanation.                                                                                                                                       |
| geographic_information                 | Information on various geographies used in the dataset provided for convenience. Includes map files showing the shapes of the geographies (states, PUMAs) used for partitioning and a lookup table mapping between census tracts and various other geographies.                                                                             |
| metadata_and_annual_results            | Building characteristics (age, area, HVAC system type, etc.) and annual results for each building energy models.                                                                                                                                                                                                                            |
| metadata_and_annual_results_aggregates | Building characteristics (age, area, HVAC system type, etc.) and annual results are aggregated to a specific geography (e.g. state, county, PUMA). Aggregate files combine duplicate models from the reallocation process within each geography, so each building ID appears only once with a weight equal to the sum of all its instances. |
| timeseries_aggregates                  | Aggregate end-use load profiles by building type and geography that can be opened and analyzed in Excel, python, or other common data analysis tools.                                                                                                                                                                                       |
| timeseries_individual_buildings        | The raw individual building timeseries data. The file names in this directory correspond to the “bldg_id” column in the metadata_and_annual_results and metadata_and_annual_results_aggregates files.                                                                                                                                       |
| weather                                | Key weather data parameters in csv format used as an input to run the building energy models to create the dataset.                                                                                                                                                                                                                         |
| data_dictionary.tsv                    | Describes the column names found in the metadata and timeseries data files. This file also specifies which columns are included in the “basic” and “full” metadata files.                                                                                                                                                                   |
| enumeration_dictionary.tsv             | Expands the definitions of the enumerations used in the metadata files.                                                                                                                                                                                                                                                                     |
| upgrades_lookup.json                   | Lookup table with upgrade ID and upgrade name for the given dataset release.                                                                                                                                                                                                                                                                |
| measure_name_crosswalk.csv             | Relates a universal “Measure ID” and the upgrade IDs and upgrade names (found in upgrades_lookup.json) across dataset releases.                                                                                                                                                                                                             |

## Metadata and Annual Results File Differences

The ComStock sampling methodology was updated as of 2024 Release 2 (see [new sampling method overview](#) for more detail). In the new sampling method, a single energy model may be used to represent multiple similar buildings across distinct but comparable geographies through a reallocation process. As a result, there are two types of metadata and annual results files available on OEDI: (1) aggregate files (in **metadata\_and\_annual\_results\_aggregates**) that combine duplicate building IDs for a given geography, and (2) non-aggregate files (in **metadata\_and\_annual\_results**) that show all instances of each building ID after the reallocation process.

Files in **metadata\_and\_annual\_results** use the higher-resolution sampling method, retaining all columns, including tract-level attributes like Cambium grid region, ISO/RTO region, and CEJST designation, for detailed regional analysis. Files in **metadata\_and\_annual\_results\_aggregates** consolidate duplicate models within each geography and omit columns that cannot be meaningfully aggregated, providing a simpler format suited to broader, less detailed analyses.

## File Types and How to Open

The following table lists the file types included on the ComStock OEDI data lake and provides details for how to open each file type.

| File type                          | Description                                                                                                                                                                                                         | How to Open                                                                                                                                                                                               |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| .csv (Comma-Separated Values)      | A plain text file that stores tabular data, where each line is a data row and values are separated by commas.                                                                                                       | Use spreadsheet software like Excel or Google Sheets, or programmatically with tools like Python's "pandas."                                                                                              |
| .json (JavaScript Object Notation) | A lightweight, text-based format for storing structured data as key-value pairs; widely used in APIs and configurations.                                                                                            | Open with any text editor or view/edit with code editors like VisualStudioCode; programmatically accessible with Python (json module) or JavaScript.                                                      |
| .osm (OpenStudio Model)            | A file format used by OpenStudio to define energy models for buildings, including geometry, HVAC, schedules, and loads.                                                                                             | Open with the <a href="#">OpenStudio</a> Application or edit programmatically with the <a href="#">OpenStudio SDK</a> .<br>Note that OpenStudio files can be converted to EnergyPlus IDF files as needed. |
| .parquet                           | A columnar storage file format optimized for large-scale data processing, commonly used with big data tools. The file sizes are much smaller than the .csv versions and are therefore easier to download and store. | Use Python libraries like "pyarrow" or "pandas", or tools like Apache Spark.                                                                                                                              |
| .tsv (Tab-Separated Values)        | Similar to CSV, but uses tabs instead of commas to separate values; used for cleanly formatted tabular data.                                                                                                        | Open with Excel, Google Sheets (specifying tab as delimiter), or programmatically using tools such as Python.                                                                                             |

## Dataset Naming Convention

ComStock releases use the following naming convention.

<dataset type>\_<weather data>\_<year of publication>\_release\_<release number>

example: comstock\_amy2018\_2021\_release\_1

- dataset type
  - resstock = residential building stock
  - comstock = commercial building stock
- weather data
  - amy2018 = actual meteorological year 2018 (2018 weather data from NOAA ISD, NSRDB, and MesoWest)
  - tmy3 = typical weather from 1991-2005 (see [this publication](#) for details)
- year of publication
  - 2021 = dataset was published in 2021
  - 2022 = dataset was published in 2022
  - etc.
- release
  - release\_1 = first release of the dataset during the year of publication
  - release\_2 = second release of the dataset during the year of publication
  - etc.

## Major Updates to ComStock Baseline

New dataset releases include updates to the ComStock baseline model. Major updates in 2025 Release 3 are highlighted below. For a complete list of baseline changes in this release, see the [GitHub release change log](#).

- **New building type: Grocery stores**  
ComStock now models grocery store buildings across the U.S. The building type metadata, annual energy, load profiles, and measure results are integrated into the ComStock dataset release under the building type name “Grocery.” For modeling details, please see the [ComStock Reference Documentation](#).

## New Upgrade Measures

New dataset releases include new upgrade measures as well as all existing measures from the previous releases. The following table summarizes the new upgrade measures introduced in this dataset release. For a full list of available measures and links to detailed documentation, please see the [Upgrade Measures](#) page on the ComStock user resource website.

| Measure ID | Measure Name                 | Description                                                                                                                                                                                                                     |
|------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hvac_0028  | High-Efficiency Rooftop Unit | This measure models the replacement of existing rooftop units (RTUs) across the U.S. commercial building stock with high-efficiency models. The term “high-efficiency” refers to top-performing RTUs currently available on the |

| Measure ID | Measure Name                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |                                                     | U.S. market as of July 2025. These systems provide cooling through direct expansion (DX) units and heating through either gas furnaces or electric resistance, consistent with each building's existing heating source.                                                                                                                                                                                                                                                                                                                           |
| hvac_0029  | Variable Speed Pumps                                | This measure replaces existing pump motors in water-based heating and cooling systems with high-efficiency variable-speed pumps, excluding those used for domestic water. The term "high-efficiency" refers to top-tier products currently available on the U.S. market as of July 2025. The pump motors are typically synchronous reluctance motors, which operate at synchronous speed by generating torque through the magnetic reluctance difference in the rotor's anisotropic paths, without requiring permanent magnets or rotor windings. |
| hvac_0030  | Thermostat Setbacks During Unoccupied Periods       | This measure models thermostat setbacks during unoccupied periods in zones that do not already have them. The setbacks adjust temperatures by 10°F for heating and 5°F for cooling, with an optimum start sequence beginning three hours before occupancy. Additionally, unoccupied heating and cooling setpoints are limited to a minimum of 55°F and a maximum of 82°F.                                                                                                                                                                         |
| hvac_0031  | Fan Static Pressure Reset for Multizone VAV Systems | This measure applies a static pressure reset for air-handling unit fans in multizone VAV systems, reducing fan energy by allowing lower pressure operation during part-load conditions. It simulates this effect by adjusting the fan power curve based on airflow, replacing the existing curve with one that reflects an effective static pressure reset. The chosen curve is used in OpenStudio Standards and is documented in the California Energy Commission's <a href="#">Advanced VAV Design Guide</a> .                                  |

## Known Issues

There is one known issue with this dataset release, summarized below.

- In ComStock 2025 Release 3, service water heating (SWH) was not modeled in any California buildings. SWH fields will show zero energy, even when we expect the model to have SWH loads. See the [known issue documentation](#) for more information.

## Citation and Data Attribution

### Suggested Citation

Parker, Andrew, et al. 2023. ComStock Reference Documentation. Golden, CO: National Renewable Energy Laboratory. NREL/TP-5500-83819.  
<https://www.nrel.gov/docs/fy23osti/83819.pdf>

## Data Attribution

When using ComStock data in products or reports, kindly use the suggested citation above or “Data includes information from the ComStock™ dataset developed by the National Laboratory of the Rockies (NLR) with funding from the U.S. Department of Energy (DOE).”

## Disclaimer

This disclaimer applies to the ComStock Dataset (“Data”). This Data was created by Alliance for Energy Innovation, LLC (“Alliance”), the manager and operator of the National Laboratory of the Rockies under Contract No. DE-AC36-08GO28308 with the U.S. Department of Energy. This Data is provided for research purposes only, and Alliance has no obligation to update, confirm the accuracy of, or in any way maintain the Data. Any use of or reliance on the Data is at your own risk. THIS DATA IS PROVIDED BY ALLIANCE “AS IS” AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL ALLIANCE OR CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS DATA, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.